

DRAFT

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Question #: 1

A 25-year-old man comes to the physician because of a 2-year history of progressively worsening myoclonic epilepsy. Laboratory testing shows elevated blood and CSF lactate levels. He has a family history as follows: his three siblings all have chronic muscle weakness, a maternal uncle with a myopathic disorder; a maternal aunt with progressive dementia who developed ataxia at 37 years; and an 80-year-old maternal grandmother with a life long history of deafness, diabetes, and renal dysfunction. Physical exam shows generalized muscle wasting and weakness, myoclonus, ataxia, and neurosensory hearing loss. He has slow nerve conduction velocities and electroencephalogram (EEG) shows bursts of slow wave and spike complexes. Muscle biopsy identifies abnormal mitochondria, deficient staining for cytochrome oxidase, and ragged-red fibers (muscle fibers with subsarcolemmal mitochondria that stains red with Gomori trichrome stain). Which of the following is the best probability estimate that the children of this patient will inherit the disorder?

- A. 0%
- B. 25%
- C. 33%
- D. 50%
- E. 67%
- F. 75%
- G. 100%

Question #: 2

A 1-year-old boy is brought to the physician by his parents because of a 3-month history of regression of previously attained developmental milestones. He was sitting when he was 6-months old, however, he now is unable to sit without support. Ophthalmoscopic examination shows a reddish area at the center of macula and a provisional diagnosis of Tay-Sachs disease is made. Samples are obtained for ASO blot analysis for the most common pathogenic variants of the *HEXA* gene. Which of the following is most likely required to perform this technique?

- A. Enzyme linked antibodies
- B. DNA polymerase
- C. SDS polyacrylamide gel electrophoresis
- D. Fluorescently labeled dideoxynucleotides
- E. A labeled single stranded oligonucleotide probe

Question #: 3

A 5-year-old boy is brought to the physician by his mother because of mild intellectual disability, facial anomalies, and a high-pitched nasal voice. He was born with a heart defect, and has had frequent infections. Due to these symptoms, 22q11.2 DS (DiGeorge Syndrome) is suspected. A G-band metaphase karyotype is equivocal, and the direct test of the fragile-X triplet repeat is negative. The most important genetic test for this child would be microarray CGH, but it is not currently offered where this patient is. Which of these other tests would most likely be able to directly test if this patient has 22q11.2 DS?

- A. Sizing of PCR products
- B. Fluorescence in situ hybridization
- C. Southern blot
- D. ASO blot
- E. cDNA microarray

Question #: 4

A 3-month-old girl is brought to the physician by her parents because of congenital malformations including microcephaly, hypertelorism (wide spaced eyes), micrognathia (small jaw), and low set and posteriorly rotated ears. At birth, she was noted to have hypotonia and a characteristic cat-like cry. Genomic testing shows that she has a 2.5 Mb deletion of chromosome 5p and she is diagnosed with Cri du Chat syndrome. Which of the following genetic tests was most likely used to obtain this diagnosis?

- A. G-band karyotype
- B. Spectral karyotype
- C. cDNA microarray
- D. Southern blot
- E. Microarray CGH

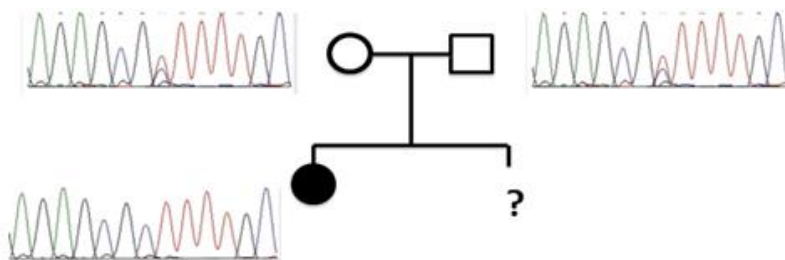
- F. PCR RFLP
- G. ASO Blot

Question #: 5

A 3-year-old girl of consanguineous parents is brought to the physician because of developmental delay, autistic behavior, cleft lip, ptosis, and cupped ears. Proband only exome sequencing shows that she has a pathogenic mutation that explains her clinical features. Sanger sequencing of the patient and her parents verifies that she has the mutation as shown in the figure. Which of the following would be the best estimate of the probability that these parents will have a second child with the disorder?

- A. 100%
- B. 75%
- C. 67%
- D. 50%
- E. 33%
- F. 25%
- G. 0%

Attachment:



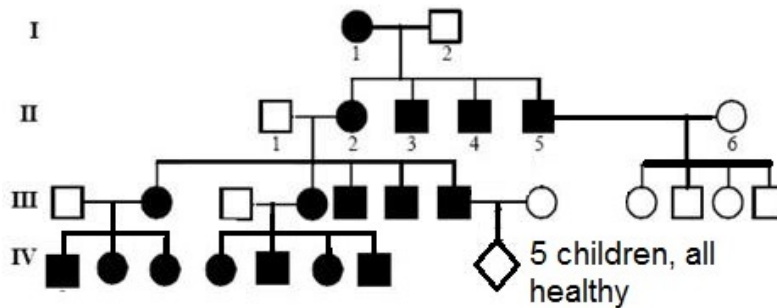
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Question #: 6

A 37-year-old woman (individual III-8 of the pedigree) comes to a geneticist to learn more about the disorder on her husband's side of the family. The features of the affected family members vary widely and include muscle twitches, weakness, and progressive stiffness. Some of the affected members display seizures and others display some degree of ataxia. One family member has developed hearing loss, and another loss of visual acuity. A pedigree of the family is shown. Which of the following disorders is likely found in this family?

- A. Becker muscular dystrophy
- B. Myoclonic epilepsy with ragged red fibers
- C. Prader- Willi syndrome
- D. Marfan syndrome
- E. Lesch-Nyhan syndrome

Attachment:



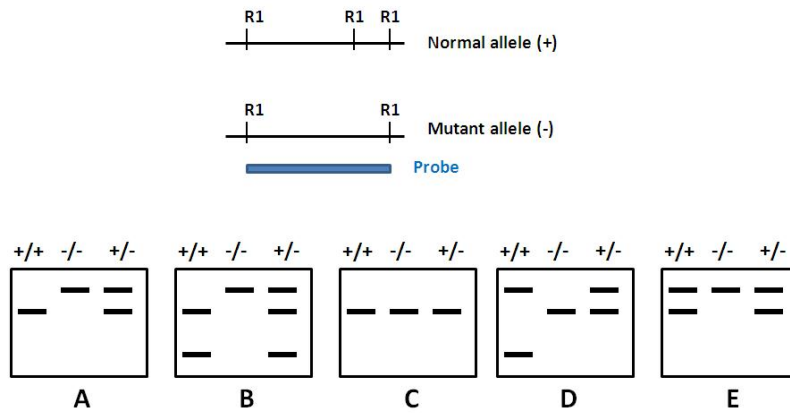
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Question #: 7

A researcher is performing RFLP analysis on the normal and mutant alleles of a specific gene in the population. The figure shows the location of *EcoRI* restriction endonuclease recognition sites (labeled as R1) for the two alleles (normal and mutant) of the gene. The shaded bar indicates the relative location of a DNA fragment that is used as a probe in the RFLP analysis. DNA samples from three individuals are analyzed: homozygous normal (+/+); homozygous mutant (-/-) and a heterozygous individual (+/-). The DNA from these individuals is digested with the *EcoRI* enzyme and the RFLP results are shown. Which of the following results would most likely be expected from this analysis?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



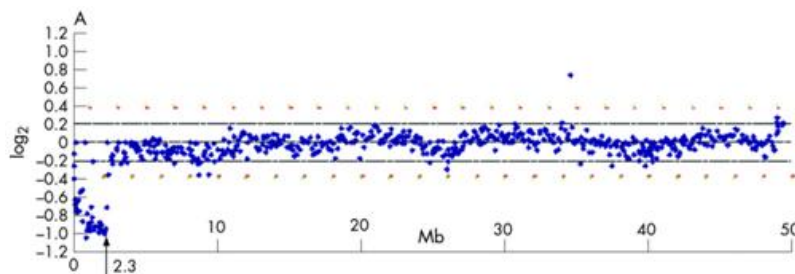
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Question #: 8

A 13-year-old boy is brought to the physician by his mother because of intellectual impairment and developmental delay. Microarray comparative genome hybridization of his genome shows an alteration on chromosome 4 as shown in the figure. Which of the following is the best interpretation of these data?

- A. Approximate two megabase deletion on chromosome four
- B. Approximate two megabase insertion on chromosome four
- C. Approximate two megabase duplication on chromosome four
- D. Overexpression of genes on chromosome four
- E. Creation of a dicentric locus on chromosome four

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Question #: 9

A 50-year-old man comes to the physician for a follow-up examination. Two months earlier he was diagnosed with familial cardiomyopathy. Targeted gene mutation analysis of the α -tropomyosin gene is performed. DNA and RNA are collected and Southern and Northern blot

analyses are performed. The data is shown.

- FIGURE A: *EcoRI* restriction map of the human α -tropomyosin gene. Exons are indicated as shaded boxes, and introns are indicated as open/non-shaded boxes. The location of the *EcoRI* restriction sites are indicated by an asterisk (*), and the distance between these restriction sites is also indicated. The location of the ^{32}P -labeled probed used for the Southern and Northern blot is shown.

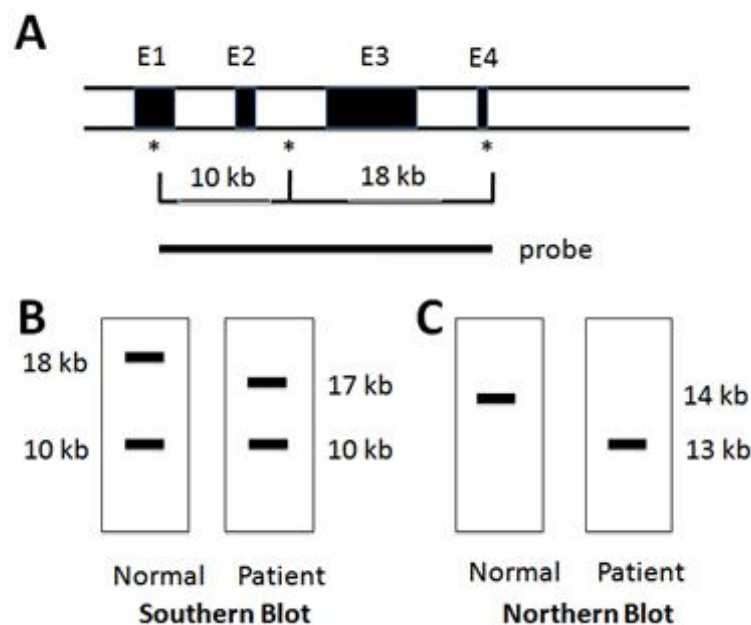
- FIGURE B: Southern blot

- FIGURE C: Northern blot

Which of the following is the most likely conclusion from the laboratory data?

- A. There is a deletion of a 1 kb segment of DNA in Exon 3
- B. The sequence of the α -tropomyosin protein is normal
- C. There is a single base mutation that affects a splicing site
- D. The α -tropomyosin gene is not transcribed
- E. There is a deletion of a 1 kb segment of DNA in Intron 3

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20665.JPG

Question #: 10

A 7-year-old girl is brought to the physician by his parents for a follow-up examination because of developmental delay. Physical examination shows that she is on the 90th percentile for weight and 3rd percentile for height. A picture of the patient is attached. Genetic analysis shows maternal uniparental disomy of

chromosome 15. Which of the following syndromes does this patient most likely have?

- A. Angelman syndrome
- B. Rett syndrome
- C. Down syndrome
- D. Prader Willi syndrome
- E. Beckwith Wiedemann syndrome

Attachment:



Girl with PW.JPG

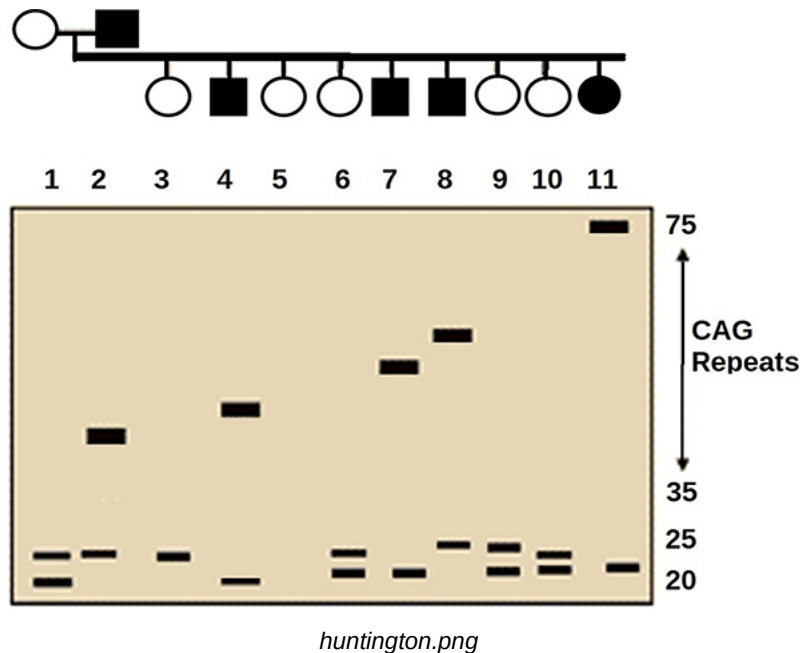
Question #: 11

A 42-year-old woman is brought to the physician by her husband because of a 6-month history of early onset dementia and chorea. Her father also developed similar symptoms when he was in his late sixties. A region within the first exon of a candidate gene was analyzed from all members of the immediate family. The attached figure shows the family pedigree, filled in symbols represent affected individuals, and the gel electrophoresis below each symbol shows

the results of the genetic test for each person. Based on these data, which of the following conclusions can most likely be drawn regarding this neurodegenerative disorder?

- A. The disorder is X-linked dominant
- B. The disorder is X-linked recessive
- C. Individual 11 will have the earliest onset age
- D. Definitive diagnosis requires karyotype
- E. Individual 1 (the mother) must be a carrier

Attachment:



Question #: 12

A 2-year-old girl is brought to the pediatric clinic by her parents due to concerns about her developmental progress. They report that she has significant delays in speech and motor skills. She often exhibits a happy demeanor with frequent smiling and laughter. On physical examination, she has a wide mouth, and widely spaced teeth. Her movements are jerky and uncoordinated. Genetic testing is ordered for the patient and her parents, often referred to as "Trio Testing". The results show that the patient has a deletion on the maternal chromosome 15q11-q13. Which of the following is the most likely diagnosis?

- A. Angelman syndrome
- B. Rett syndrome

- C. Down syndrome
- D. Prader Willi syndrome
- E. Fragile X syndrome

Question #: 13

A 4-year-old boy is brought to the physician due to a 1-year history of global developmental delays and facial dysmorphologies. His parents report that he has had multiple congenital anomalies from birth. Physical examination shows a broad forehead, hypertelorism (closely spaced eyes), and a small chin. The physician orders both a SNP and CGH microarray to investigate the underlying genetic cause of the features seen in the patient. Which of the following best describes the utility of using both the SNP and CGH microarray for this patient?

- A. They will detect single nucleotide polymorphisms (SNPs) associated with common diseases
- B. They will detect chromosomal copy number variations (CNVs) and regions of homozygosity
- C. They will give sequence information for the entire genome to identify pathogenic mutations
- D. They will detect both balanced and unbalanced chromosomal rearrangements
- E. They will identify both genomic and mitochondrial DNA mutations

Question #: 14

A researcher is investigating the intracellular mechanism of action of a hormone. She plans to develop drugs that mimic the hormone's activity. The hormone facilitates the intracellular release of sequestered Ca^{2+} from the endoplasmic reticulum. Which of the following messengers most likely performs this action?

- A. Diacylglycerol
 - B. Inositol 1, 4, 5-tris-phosphate
 - C. Phosphatidylinositol 4,5-bisphosphate
 - D. Cyclic-AMP
 - E. Cyclic-GMP
 - F. Nitric oxide
-

Question #: 15

A 62-year-old woman is brought to the emergency department because of a 4-hour history of crushing chest pain that radiates to her left arm. Laboratory studies show elevated serum troponin-T levels. Which of the following is the most likely diagnosis?

- A. Acute pancreatitis
- B. Muscular dystrophy
- C. Myocardial infarction
- D. Acute hepatocellular damage
- E. Obstructive jaundice

Question #: 16

A 7-month old boy is brought to the physician by his mother because of a failure to thrive and poor weight gain since birth. Physical examination shows hepatomegaly and jaundice. There is no evidence of biliary obstruction. Which of the following serum marker levels is most likely elevated in this patient?

- A. Amylase
- B. Lipase
- C. Alanine aminotransferase
- D. Alkaline phosphatase
- E. Creatine kinase-MB
- F. Troponin-T

Question #: 17

A researcher has identified a drug 'X' which inhibits the G protein-coupled receptor-stimulated phospholipase C activity in intact cell systems. The intracellular changes in signal transduction are observed before and after administration of the drug. Which of the following intracellular changes would most likely be observed following administration of the drug?

- A. Decreased intracellular cAMP levels
- B. Increased intracellular levels of inositol trisphosphate (IP3)
- C. Decreased cytosolic calcium levels
- D. Increased protein kinase C activity

- E. Increased intracellular levels of diacylglycerol
- F. Decreased activity of protein kinase A
- G. Decreased activity of guanylate cyclase

Question #: 18

A 10-year-old boy is brought to the emergency department by his mother because of a sudden-onset of nausea, vomiting and severe respiratory distress. His mother says that he accidentally spilled a bottle of pesticide (containing organophosphates, DIFP). Ophthalmic examination shows pupillary constriction (miosis). Which of the following is the most likely effect of this toxin on acetylcholinesterase?

- A. Increases the apparent K_m
- B. Binds covalently to the active site
- C. Binds reversibly to a sulfhydryl group
- D. Acts as a feedback inhibitor
- E. Causes reversible inhibition
- F. V_{max} is unchanged

Question #: 19

A 67-year-old man comes to the physician for a follow-up examination. During a previous visit 3-months ago, laboratory studies showed his serum LDL cholesterol was 280mg/dL (Normal: <160mg/dL). He was prescribed statins which act as enzyme inhibitors. Which of the following enzyme kinetic changes most likely occurs in the presence of this inhibitor?

- A. The K_m and V_{max} are unchanged
- B. The apparent K_m is increased but V_{max} is the same
- C. The apparent K_m and apparent V_{max} are decreased
- D. The K_m is unchanged but apparent V_{max} is decreased
- E. Statin binds irreversibly to the active site of the enzyme

Question #: 20

An investigator is studying the effect of nitric oxide on blood vessels and observes an increase in the intracellular concentration of a specific second messenger in the smooth muscle cells in the walls of the blood vessels. Which of the following enzymes is most likely activated in the smooth

muscle cells to produce this second messenger?

- A. Guanylate cyclase
- B. Adenylate cyclase
- C. Phosphodiesterase
- D. Phospholipase C
- E. Protein kinase A
- F. Protein kinase C

Question #: 21

A 30-year-old woman comes to the physician because of a 2-day history of severe watery diarrhea and vomiting. Physical examination shows shrunken skin and dehydration. Stool tests show *Vibrio cholerae*. Which of these modifications is most likely responsible for the pathology in this patient?

- A. ADP ribosylation of the G α subunit
- B. ADP ribosylation of the G β subunit
- C. ADP ribosylation of the CFTR protein
- D. ADP ribosylation of the EF-2
- E. ADP ribosylation of the G γ subunit

Question #: 22

A 55-year-old man is brought to the emergency room because of a 3-hour history of worsening chest pain radiating down the left arm. The pain began following a game of tennis 3-hours ago. The intensity of the pain has increased progressively and is accompanied by profuse sweating. An ECG is inconclusive, but laboratory tests show myocardial infarction. Which of the following laboratory results best supports the diagnosis?

- A. Elevated serum activities of creatine kinase-BB (CK-BB)
 - B. Elevated serum activities of CK-2 and total CK
 - C. Increased serum alkaline phosphatase levels
 - D. Increased serum amylase and lipase levels
 - E. Elevated serum alanine aminotransferase levels
-

Question #: 23

A 16-month-old boy is brought to the physician by his parents for a follow up examination. He has had repeated episodes of respiratory infections in the past two months. Laboratory studies show elevated sweat chloride levels. Genetic tests show that the child is homozygous for the most common mutation in the *CFTR* gene. Which of the following best describes the mutation causing this pathology?

- A. Deletion of a single amino acid in CFTR protein
- B. Deletion of 10 base pairs in the CFTR gene
- C. Deletion of the entire CFTR gene
- D. A glutamate is replaced by valine in the CFTR protein
- E. A frame shift mutation in the CFTR gene
- F. The CFTR protein is shorter by three amino acids

Question #: 24

A 22-year-old-man comes to the physician for a follow up examination regarding a genetic disorder that was diagnosed a month ago. He has a 3-month history of proximal muscle weakness. Genetic studies show that he has an in-frame deletion in the *DMD* gene which encodes a protein essential for maintaining muscle fiber integrity. Which of the following laboratory findings would most likely be observed in this patient?

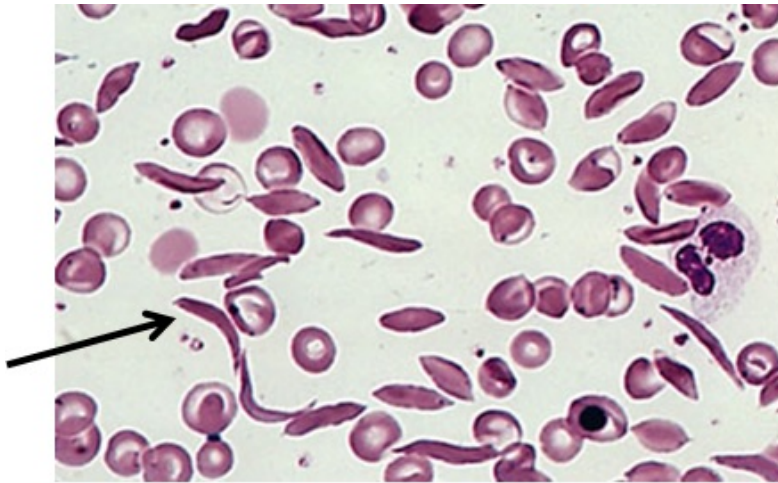
- A. Elevated chloride levels in sweat
- B. Reduced staining for dystrophin in muscle biopsy
- C. Absence of dystrophin staining on muscle biopsy
- D. Increased serum creatine kinase-BB levels
- E. Anemia and decreased hemoglobin levels

Question #: 25

A 3-year-old girl is brought to the physician because of a 6-hour history of intense muscular pain after exercise. She has an older brother who was diagnosed with a genetic disorder. Physical examination shows pallor and generalized pain. A peripheral blood smear is shown. Which of the following molecular changes best explains the clinical presentation in this patient?

- A. A nonsense mutation in the beta-globin gene
- B. A hydrophobic pocket forms on the surface of Hb
- C. A glutamate to phenylalanine substitution in beta-globin chains
- D. A frameshift mutation in the beta-globin gene
- E. Point mutation in the alpha-globin gene of hemoglobin

Attachment:



170033.jpg

Question #: 26

A 49-year-old man comes to the physician because of a 3-month history of loss of appetite, cough and shortness of breath. He has smoked 5-pack year history of cigarette smoking. Physical examination shows hoarseness of voice (dysphonia), emaciation (abnormally thin) and decreased muscle strength. A biopsy specimen obtained from the bronchi via bronchoscopy shows stratified squamous non-keratinized epithelium (abnormal). Following successful treatment, a repeat biopsy shows normal bronchial respiratory epithelium. Which of the following best describes this change?

- A. Dysplasia
- B. Neoplasia
- C. Metaplasia
- D. Hypertrophy
- E. Atrophy

Question #: 27

A 12-year-old boy is brought to the emergency department because of a 1-day history of painful, itchy "sore" on his right leg, nausea, and muscle and joint pain. He received a spider bite at the same site 2 days ago. Physical examination shows a fluid-filled blister that is 2-cm in diameter, with a red margin. He is diagnosed with a necrotizing ulcer from the venomous bite of a spider. Which of the following best characterizes this type of cell death?

- A. Plasma membrane blebbing
- B. Preservation of homeostasis
- C. Cell swelling
- D. Chromatin aggregation
- E. DNA fragmentation

Question #: 28

A 45-year-old woman comes to the physician because of a 1-week history of abdominal pain. History shows she frequently drinks alcohol (10 bottles/week). Physical examination shows yellowing of the sclera with shaking of the outstretched hands. A liver biopsy is collected and shows fatty change with accumulation of triglycerides in the hepatocytes' cytoplasm. Which of the following best describes the changes observed in this patient?

- A. Dysplasia
- B. Early cellular injury
- C. Atrophy
- D. Irreversible cellular injury
- E. Neoplasia

Question #: 29

A 1-month-old boy who was born in a rural clinic is brought to the Pediatric clinic by his parents for review. He was born with a musculoskeletal abnormality of the right foot. Physical examination shows fusion of the right 1st and 2nd toes at the 1st web space. He is diagnosed with Syndactyly (failure of digits to separate into individual appendages due to inadequate programmed cell death during fetal development). Which of the following best characterizes the

defective type of cell death causing this patient's congenital malformation?

- A. Inflammation
- B. Lack of homeostasis
- C. Cell swelling
- D. Chromatin aggregation
- E. Inflammation

Question #: 30

A 73-year-old woman comes to the physician for a routine examination. She has a history of right breast cancer for which she has completed treatment by radiation therapy 6 weeks ago. Some of the radiation doses were delivered to normal surrounding tissues, resulting in cell injury and the formation of a cystic mass. Tissue collections of the cyst were obtained by surgical resections. Western blot analysis of the cells shows elevated levels of the Bcl-2-associated X (BAX) protein. Which of the following will these cells most likely exhibit?

- A. Release of cytochrome C from mitochondria
- B. Extracellular lactate dehydrogenase
- C. Random DNA degradation
- D. Cell swelling
- E. Positive propidium iodide staining